# Siyuan Chang

Email: csy24@ruc.edu.cn | Phone: +86 17372238022
Renmin University of China | Homepage: https://SummerChangDog.github.io/

## Education

**Renmin University of China, School of Statistics** Sep 2024 – Present
Data Science and Big Data Technology
• **GPA:** 3.9/4.0, Top 5% in major
• **Selected Coursework:** Machine Learning Theory; Advanced Optimization; High-Dimensional Statistics; Probability Theory; Stochastic Processes; Reinforcement Learning; Deep Learning; Parallel Computing; Data Structures.
• **Language:** CET-4: 688, CET-6: 656;second foreign language: French
• **Honor:** First-class Academic Excellence Scholarship, RUC, 2025 & 2026

## Academic Papers

[1] WED-Net: A Weather-Effect Disentanglement Network with Causal Augmentation for Urban Flow Prediction.
Qian Hong, **Siyuan Chang**, Xiao Zhou.
*WWW 2026*. Paper Link.
[2] Controllable Affective Generation via Latent Vector Steering.
Xixian Yong, **Siyuan Chang**, Xiao Zhou, Zihe Wang, Yingying Zhang,Yefeng Zheng, Xian Wu.
*EMNLP 2026*. Paper Link.

## Research Experiences

**Bandit Convex Optimization with Gradient Variation** Sep. 2026 – Present
*Advisor: Peng Zhao*

- Investigate high-probability regret bounds for two-point BCO against adaptive adversaries.
- Extend gradient-variation-based bandit learning to bilinear games.
- Explore improved dimension dependence for regret bounds under convex regret.

**Controllable Affective Generation via Latent Vector Steering** Sep. 2025 – Dec. 2025
*Advisor: Xiao Zhou*

- Propose EmoVec, a post-hoc framework for controllable affective generation in LLMs via latent vector steering, addressing emotional flattening caused by alignment techniques without weight tuning or retraining.

**WED-Net: A Weather-Effect Disentanglement Network with Causal Augmentation for Urban Flow Prediction** Jun. 2025 – Aug. 2025
*Advisor: Xiao Zhou*

- Disentangle weather-induced traffic patterns via attention mechanisms, with a weather discriminator and a causal data augmentation strategy that preserves causal structures while perturbing non-causal components.

## Research Interests

**Online Learning**: Developing algorithms that remain effective and computationally efficient over long horizons, even under uncertainty and limited information.

- **Game Theory**: No-regret learning and equilibrium convergence in multi-agent systems, with a focus on learning dynamics, gradient variation, and parameter-free algorithms.
- **Bandit Learning**: Regret minimization under limited feedback, focusing on the exploration–exploitation tradeoff and obtaining regret guarantees against both oblivious and adaptive adversaries.
- **Reinforcement Learning**: Sequential decision-making under uncertainty, with a particular interest in its fundamental connections to online learning, game theory, and bandit learning.